

Dietary protein and bone health: harmonizing conflicting theories.

A precise understanding of the role of dietary protein in bone health has been evasive despite decades of research. It is known that a dietary acid load is harmful to bone, and sulfur-containing amino acids are metabolized to provide such an acid load. It is also known that protein elevates urine calcium loss. However, recent clinical studies and a meta-analysis have indicated either no effect or a modest benefit associated with higher protein intakes. These contradictory considerations may be explained by the existence of a two-faced relationship between protein and bone, with simultaneous positive and negative pathways. In opposition to the negative effects of dietary acid load, protein may exert positive effects related to improving calcium absorption, increasing insulin-like growth factor 1, or improving lean body mass, which, in turn, improves bone strength. Putative mechanisms behind these pathways are reviewed here, and some limitations in the historical literature as well as suggested measures to counter these in the future are identified. When positive and negative pathways are considered in tandem, protein may offer modest benefits to bone in the presence of adequate dietary calcium and acid-neutralizing fruits and vegetables. © 2011 International Life Sciences Institute.